# DOBUTEL Concentrate for Solution for Infusion 50mg/ml

Dobutamine 50mg/ml

**Product Description**

Before and after dilution: Clear solution, colourless to yellowish, practically free from visible particles.
The product is not supplied with the diluents.

**Composition**

Each vial of 5ml contains a total of Dobutamine Hydrochloride equivalent to Dobutamine 250mg.

**Pharmacodynamics**

DOBUTEL is a direct-acting inotropic agent whose primary activity results from stimulation of the β-receptors of the heart while producing comparatively mild chronotropic, hypertensive, arrhythmogenic, and vasodilative effects. It does not cause the release of endogenous norepinephrine, as does dopamine. In animal studies, dobutamine hydrochloride produces less increase in heart rate and less decrease in peripheral vascular resistance for a given inotropic effect than does isoproterenol.

**Pharmacokinetics**

In patients with depressed cardiac function, both dobutamine and isoproterenol increase the cardiac output to a similar degree. In the case of dobutamine, this increase usually not accompanied by marked increases in heart rate and cardiac stroke volume is usually increased. Facilitation of atrioventricular conduction has been observed in human electrophysiologic studies and in patients with atrial fibrillation. Systemic vascular resistance is usually decreased with administration of dobutamine. Occasionally, minimum vasoconstriction has been observed. The onset of action of DOBUTEL is within 1 to 2 minutes; however, as much as 10 minutes may be required to obtain the peak effect of a particular infusion rate. The plasma half-life of dobutamine in humans is 2 minutes. The principal routes of metabolism are methylation of the cathechol and conjugation. In human urine, the major excretion products are the conjugates of dobutamine and 3-O-methyl dobutamine. The 3-O-methyl derivative is inactive.

**Indication**

Dobutamine Hydrochloride is indicated in adults who require short-term treatment of cardiac failure secondary to acute myocardial infarction or cardiac surgery.

**Recommended Dosage**

*Administration*
Because of its short half-life, Dobutamine hydrochloride must be administered as a continuous intravenous infusion. This is most reliably accomplished using a precision volume control intravenous infusion set. Following the initiation of a constant rate infusion or upon changing the rate, a steady-state dobutamine plasma concentration is achieved within approximately 10 minutes. Thus, loading doses or bolus injections are not necessary are not recommended.

*Dosage unit*
Most reports on Dobutamine hydrochloride have expressed the dose in relation to body mass, for example, mcg/kg/min. This practice is useful to relate doses in infants and children to those in adults. Among adults, body mass has little influence on the effect of Dobutamine hydrochloride; since the dosage should be titrated in each patient, adults may be easily dosed with mcg/min units. The dosage of Dobutamine hydrochloride may be initiated at 100-200 mcg/min and increased gradually to 1000-2000 mcg/min or greater, depending on the clinical and hemodynamic response of the individual patient.

DOBUTEL must be diluted to at least 50mL of compatible IV solution at the time of administration.

Recommended Dosage:
The rate of infusion needed to increase cardiac output usually ranges from 2.5-10 µg/kg/min (see table). Frequently, doses up to 20µg/kg/min are required for adequate hemodynamic improvement. On rare occasions, infusion rates up to 40 µg/kg/min have been reported.

***Rate of Infusion for Concentration of 250, 500, and 1000 mcg/mL***

| Drug Delivery Rate µg/kg/min | 250 mcg/mL* mL/kg/min | 500 mcg/mL** mL/kg/min | 1000 mcg/mL*** mL/kg/min |
|---|---|---|---|
| 2.5 | 0.01 | 0.005 | 0.0025 |
| 5 | 0.02 | 0.01 | 0.005 |
| 7.5 | 0.03 | 0.015 | 0.0075 |
| 10 | 0.04 | 0.02 | 0.01 |
| 12.5 | 0.05 | 0.025 | 0.0125 |
| 15.0 | 0.06 | 0.03 | 0.015 |

*250mcg/ml of admixture (add one 5 mL vial of 50 mg/mL of Dobutamine Injection to obtain a final admixture volume of 1000mL).
**500mcg/mL or 250 mg/500 mL of admixture (add one 5 mL vial of 50 mg/mL of Dobutamine Injection to obtain a final admixture volume of 500 mL).
***1,000 mcg/mL or 250 mg/250 mL of admixture (add one 5 mL vial of 50 mg/mL of Dobutamine Injection to obtain a final admixture volume of 250 mL).

The rate of administration and the duration of therapy should be adjusted according to the patient's response, as determined by the following clinical indicators: haemodynamic parameters, such as heart rate and rhythm, arterial pressure, and whenever possible, cardiac output and measurements of ventricular filling pressures (central venous, pulmonary capillary wedge and left atrial), and signs of pulmonary congestion and organ perfusion (urine flow, skin temperature and mental status).

Concentrations up to 5000 mg/L have been administered to humans (250 mg/50 mL). The final volume administered should be determined by the fluid requirements of the patient. Rather than abruptly discontinuing therapy with dobutamine hydrochloride, it is often advisable to decrease the dosage gradually.

Compatibilities:
DOBUTEL is compatible with the following infusion solutions:
a) 0.9% Sodium Chloride
b) 5% Glucose
c) 5% Glucose in 0.9% Sodium Chloride
d) Lactated Ringer's

The solution will remain stable for 30 hours at room temperature (30 °C) after diluted with the compatible solutions.
Solution of Dobutamine hydrochloride may have a pink discoloration. This discoloration, which will increase with time, results from a slight oxidation of the drug. However, there is no significant loss of drug potency within the recommended storage times for solutions of the drugs.

**Route of Administration**

Intravenous infusion

**Contraindications**

· Patients with idiopathic hypertrophic subaortic stenosis.
. Patients who have shown previous manifestation of hypersensitivity to dobutamine hydrochloride.

**Warnings and Precautions**

**Precautions:**

During the administration of Dobutamine Hydrochloride, as with any adrenergic agent, ECG and blood pressure should be continuously monitored. In addition, pulmonary artery wedge pressure and cardiac output should be monitored whenever possible to aid in the safe and effective infusion of Dobutamine Hydrochloride. Hypovolemia should be corrected with suitable volume expanders before treatment with Dobutamine Hydrochloride is instituted. In patients who have atrial fibrillation with rapid ventricular response, a digitalis preparation should be used prior to instituting therapy with Dobutamine Hydrochloride. Dobutamine Hydrochloride may be ineffective if the patients has recently received a beta blocking drug. In such a case, the peripheral vascular resistance may increase. No improvement may be observed in the presence of marked mechanical obstruction, such as severe valvular aortic stenosis. There is concern that any agent which increases contractile force and heart rate may increase the size of an infraction by intensifying ischaemia, but it is not known whether dobutamine does so in man.

Pediatric Use: The safety and efficacy of Dobutamine Hydrochloride for use in children have not been studied.

Usage in Hypotension: When hypotension is largely attributable to decreased cardiac output and coincides with an elevated ventricular filling pressure, the infusion of dobutamine HCL may help to restore the pressure. As volume is replenished in the treatment of acute hypoperfusion states and there is an increase in pulmonary wedge or central venous pressure but with no increase in cardiac output and arterial pressure, dobutamine may improve output and help restore the arterial pressure. In general, when mean arterial blood pressure is 70 mmHg in the absence of an increased ventricular filling pressure, hypovolemia may be present and would require treatment with appropriate volume repleting solution before dobutamine HCL is given. If arterial blood pressure remains low or decreases progressively during administration of dobutamine HCL despite adequate ventricular filling pressure and cardiac output, consideration may be given to the concomitant use of a peripheral vasoconstrictor agent, e.g. dopamine or norepinephrine.

Reactions at Site of Intravenous Infusion: Phlebitis has occasionally been reported. Local inflammatory changes have been described following inadvertent infiltration. Isolated cases of cutaneous necrosis have been reported.

**Warnings:**

Increase in Heart Rate or Blood Pressure: Dobutamine Hydrochloride may cause a marked increase in heart rate or blood pressure, especially systolic pressure. Approximately 10 percent of patients have had rate increases of 30 beats/minute or more, and about 7.5 percent have a 50 mmHg or greater increase in systolic pressure. Reduction of dosage usually reverses these effects promptly. Because dobutamine facilitates atrioventricular conduction, patients with atrial fibrillation are at risk of developing an exaggerated pressor response.

Ectopic Activity: Dobutamine Hydrochloride may precipitate or exacerbate ventricular ectopic activity, but it rarely has caused ventricular tachycardia.

Anaesthetics: The myocardium may be sensitized to the effect of dobutamine by cyclopropane or halogenated hydrocarbon anaesthetics, and these should be avoided.

Warning for Sodium Metabisulphite: This preparation contains Sodium metabisulphite that may cause serious allergic type reactions in certain susceptible patients. Do not use if known to be hypersensitive to bisulphites.

**Interactions with Other Medicaments**

Halogenated anaesthetics:
Although it is less likely than adrenaline to cause ventricular arrhythmias, DOBUTEL should be used with great caution during anaesthesia with cyclopioprane, halothane and other halogenated anaesthetics.

Entacapone:
The effects of DOBUTEL may be enhanced by entacapone.

Beta-blockers:
The inotropic effect of dobutamine stems from stimulation of cardiac beta1 receptors, this effect is reversed by concomitant administration of beta-blockers. Dobutamine has been shown to counteract the effect of beta-blocking drugs. In therapeutic doses, dobutamine has mild alpha1- and beta2-agonist properties. Concurrent administration of a non-selective beta-blocker such as propranolol can result in elevated blood pressure, due to alpha-mediated vasoconstriction, and reflex bradycardia. Beta-blockers that also have alpha-blocking effects, such as carvedilol, may cause hypotension during concomitant use of dobutamine due to vasodilation caused by beta2 predominance.

**Fertility, Pregnancy and Lactation**

Reproduction studies in rats and rabbits have revealed no evidence of impaired fertility, harm to the foetus, or teratogenic effects due to dobutamine. As there are no adequate and well-controlled studies in pregnant women, and as animal reproduction studies are not always predictive of human response, dobutamine should not be used during pregnancy unless the potential benefits outweigh the potential risks to the foetus.

**Side Effects**

*Adult population*
Infusions for up to 72 hours have revealed no adverse effects other than those seen with shorter infusions. There is evidence that partial tolerance develops with continuous infusions of Dobutamine 25 mg/ml concentrate for solution for infusion for 72 hours or more; therefore, higher doses may be required to maintain the same effects.

Immune system disorders
*Not Known:* Hypersensitivity reactions including rash, fever, eosinophilia and bronchospasm have been reported. Anaphylactic reactions and severe life-threatening asthmatic episodes may be due to sulfite sensitivity.

Blood and lymphatic system disorders
*Common:* Eosinophilia, inhibition of thrombocyte aggregation (only when continuing infusion over a number of days)

Metabolism and nutrition disorders
*Very rare:* Hypokalaemia

Nervous system disorders
*Common:* Headache
*Not known:* Paraesthesia, tremor, myoclonic spasm. Myoclonus has been reported in patients with severe renal failure receiving dobutamine;

Cardiac disorders/vascular disorders
*Very common:* Increase of the heart rate by ≥ 30 beats/min
*Common:* Blood pressure increase of ≥ 50 mmHg. Patients suffering from arterial hypertension are more likely to have a higher blood pressure increase; Blood pressure decrease, ventricular dysrhythmia, dose-dependent ventricular extrasystoles.; Increased ventricular frequency in patients with atrial fibrillation. These patients should be digitalised prior to dobutamine infusion. Vasoconstriction in particular in patients who have previously been treated with beta receptor blockers; Anginal pain, palpitations

*Uncommon:* Ventricular tachycardia, ventricular fibrillation Very rare: Bradycardia, myocardial ischaemia, myocardial infarction, cardiac arrest
*Not known:* Electrocardiogram ST segment elevation; Decrease in pulmonary capillary pressure Eosinophilic myocarditis has been noted in explanted hearts of patients who had undergone treatment with multiple medications including dobutamine or other inotropic agents prior to transplantation.

*Children:* pronounced increase of heart rate and/or blood pressure as well as a lower decrease of the pulmonary capillary pressure than adults. Increase of pulmonary capillary pressure in children under 1.

<u>Gastrointestinal disorders</u>
*Not known:* Nausea,

<u>Psychiatric disorders</u>
Not known: Restlessness, feeling of heat and anxiety

<u>Renal and urinary disorders</u>
Not known: Urinary urgency

**Symptoms and Treatment of Overdose**

Overdoses of dobutamine have been reported rarely. The following is provided to serve as a guide if such an overdose is encountered.
*Symptoms:* Toxicity from Dobutamine hydrochloride is usually due to excessive cardiac β-receptor stimulation. The duration of action of Dobutamine hydrochloride is generally short (T1/2 = 2 min) because it is rapidly metabolised by catechol-O-methyltransferase. The symptoms of toxicity may include anorexia, nausea, vomiting, tremor, anxiety, palpitations, headache, shortness of breath and anginal and nonspecific chest pain. The positive inotropic and chronotropic effects of dobutamine on the myocardium may cause hypertension, tachyarrhythmias, myocardial ischemia and ventricular fibrillation. Hypotension may result from vasodilation. If the product is ingested, unpredictable absorption may occur from the mouth and the gastrointestinal tract.
*Treatment:* In managing overdosage, consider the possibility of multiple drug overdoses, interaction among drugs and unusual drug kinetics in the patient. The initial actions to be taken in a dobutamine hydrochloride overdose are discontinuing administration, establishing an airway and ensuring oxygenation and ventilation. Resuscitative measures should be initiated promptly. Severe ventricular tachyarrhythmias may be successfully treated with propranolol or lidocaine. Hypertension usually responds to a reduction in dose or discontinuation of therapy. Protect the patient’s airway and support ventilation and perfusion. If needed, meticulously monitor and maintain, within acceptable limits, the patient’s vital signs, blood gases, serum electrolytes, etc. Absorption of drugs from gastrointestinal tract may be decreased by giving activated charcoal, which, in many cases, is more effective than emesis or lavage, consider charcoal instead of or in addition to gastric emptying. Repeated doses of charcoal over time may hasten elimination of some drugs that have been absorbed. Safeguard the patient’s airway when employing gastric emptying or charcoal. Forced diuresis, peritoneal dialysis, haemodialysis or charcoal hemoperfusion have not been established as beneficial for an overdose of dobutamine hydrochloride.

**Effects on Ability to Drive and Use Machine**

Not applicable in view of the indications for use and the short half-life of the drug.

**Instructions for Use and Incompatibilities**

*Incompatibilities*
Do not add DOBUTEL to 5% Sodium Bicarbonate intravenous infusion BP or to any other strongly alkaline solutions. Because of potential physical incompatibilities, it is recommended that dobutamine hydrochloride not be mixed with other drugs in the same solution.

DOBUTEL should not be used with other agents or diluents containing both sodium metabisulfite and ethanol.

*Special Precautions for Disposal and Other Handling*
DOBUTEL should be diluted before use and administered by IV infusion only.
The final concentrations generally used for infusion are 250 mcg/mL, 500 mcg/mL or 1000 mcg/mL (see table *Rate of Infusion for Concentration of 250, 500, and 1000 mcg/mL*).

The following sterile solutions for IV infusion may be used for the dilution of dobutamine before use: 0.9% Sodium Chloride, 5% Glucose, 5% Glucose in 0.9% Sodium Chloride, or Lactated Ringer's

**Storage Conditions**

Before reconstituted, store below 30°C and protect from light.
After dilution with compatible infusion solution such as: 0.9% Sodium Chloride, 5% Glucose, 5% Glucose in 0.9% Sodium Chloride, and Lactated Ringer’s, the solution is stable up to 30 hours at room temperature (25 - 30ºC) and refrigerator (2 - 8°C). Protect from light.
Infusion must be aseptically prepared.
From a microbiological point of view, the product should be used immediately. If not used immediately, in-use storage times and conditions prior to use are the responsibility of the user.

**Dosage Forms and Packaging Available**

Box contains 1vial x 5ml DOBUTEL Concentrate for Solution for Infusion.
*Primary packaging:* 6ml clear vial (filled to 5ml), made from borosilicate glass (Type I) with grey bromobutyl rubber stopper (outer diameter: 18.90 - 19.20 mm) and 20mm yellow flip off seal, made of aluminium and polypropylene.
*Secondary packaging:* 1 labelled vial is packed with a package insert in a box.

**Manufacturer**

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**Product Registration Holder**

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